Sony Troubleshooting Steps

1. Connection to the Instrument and the Software
    a. Make sure the instrument on screen window states the instrument is on Standby mode
        i. Close the software and reopen after the on-screen window states standby
    b. If not connecting still turn off the instrument and restart the computer
    c. If the instrument freezes and shutdown itself in middle of the sort
        i. You must force quit the software using the task manager function and closing the program there.
        ii. Do a hard shutdown on the instrument. To do a hard shutdown you will need to hold down on the Power/Standy button on the instrument until it turns off
        iii. Wait about 30 seconds before turning on the instrument. It should reconnect to the software, but you will need to redo the full software startup and sort calibration
2. Initial instrument Setup window
    a. Step 1Chip Detection Issues
        i. When you are scanning a chip wrapper that is over 24 hours old
        ii. The computer on the camara has been “closed” If so, press on top center of the monitor and it should reveal the camara if it has been pushed down “close”
    b. Step 3 Chip loading Issues
        i. Unload the chip and double check the orientation of the chip. Should be inserting the chip reading the nozzle size left to right.
        ii. Instrument present load chip error. Just reinsert the chip and try again.
        iii. The instrument does not unload the chip
            1. You would need to shut down the main unit, turn off the air compressor supply. Pull the sample line out of the chip connector by turning the connector to the left. You will need to manual roll up the chip, the roll wheel is found next to the chip port.
    c. Step 4 Laser setting Issues
        i. When the instrument provides a error message example saying.. “Laser output error 405,561..Etc”
            1. Try turning off the instrument with a hard reset. Wait 30 seconds then try turning the instrument
            2. Still persist call the service engineer

3. Instrument should still be able to run if you do not need the failing channel. Uncheck mark the laser not functional

d. Step 5 Filter Setting Issues
    i. Should not have issues in choosing fluorescent or standard
    ii. Our instrument only has one configuration
    iii. When in doubt pick standard
e. Step 6 Fluidics Check
    i. When the droplet image in droplet viewer shakes up and down or just not still as a picture. If the flow path stream in the instrument is not center in the waste and is curved. If the shape of the droplets is irregular.
        1. In the wizard click sample line cleaning, skip bleach cleaning and run water
        2. Still not stable click chip de-bubble 3x
        3. If it persists, turn off the instrument and compressor. Check the cap of the sheath tank. Open and reset the sheath cap. Make sure the waste tank is empty starting expect for the bleach. Check if all the lines are not showing leaks and all connected.
        4. Turn on the instrument and on the login in screen, click maintenance and run Ethanol Priming to remove any air pockets
        5. If still not stable, try a new chip.

3. Automatic Calibration Wizard
    a. An error occurs in automatic calibration or automatic calibration stops and does not proceed.
        i. There may be air bubbles trapped in the sorting chip. Skip Automatic calibration, in the cytometer tab of the experiment in the de-bubble section click chip de-bubble.
        ii. Check if you place a cap or tube cover in the sample holder. If you did this will stop the instrument from pressurizing correctly.
        iii. Remake the calibration beads
    b. Step 1 Chip Alignment and Laser Delay setting Issues
        i. Note: If the status bar refreshes 3x times over on this step or error message appears. You either have old/compromise beads or the chips is damaged.
            1. Click abort, start and experiment, and redo the auto-calibration with fresh beads first, then try new chip
            2. If the error message no sample event in coarse mode. Click on the troubleshooting tab and follow the on-screen instruction to resolve the issue
    c. Step 2 Droplet Calibration Issues

i. If the status bar refreshes 3x times over
    1. Check the stability of the stream if shaking. Will need to abort and de-bubble
    2. Check the image window of the stream. If it is blurry, that means the camara to view the stream is wet and will need to call the service line to help either resolve the issue or bring in an engineer to clean the camara window. Hit abort.
    3. If error message appears. Follow the on-screen instructions to the troubleshooting mode steps

d. Step 3 Side Stream Calibration Issues
    i. If the status bar refreshed 3x times over
        1. The deflection plates could have salt build up and will need to be recleaned with a kimwipe and EtOH.
            a. To turn off the stream you will need to click abort
            b. Start an experiment
            c. In the Cytometer tab-> select settings->bottom of that window select advanced setting-> click on the pressure option tab->click standby
        2. Clean the black transparent protective covers in the collection area. It could be charged with static. Using a fresh kimwipe moistened with diH20

4. Issues when Monitoring the Sort
    a. An error message stating unable to maintain stable droplet breakoff
        i. Occurs when the control breakoff function was turned off when the stream becomes unstable for various reasons. Partial clog or clog in the instrument breakoff.
            1. If the droplet formation problem is detected during the run, acquisition will stop and will insist to run a sort calibration again.
        ii. When you notice your event rate drop to 0, even when you still have sample in the tube
            1. Try to increase the flow rate of sample. If that does not help pause the sort and resuspend/filter your sample
            2. Try a chip de-bubble. The option in in the cytometer tab under the de-bubble section select the chip icon.
            3. Enforce sample line black flush by stopping the sample. There sample could have been clog just in the sample line.
        iii. When you notice that your stream droplet viewer is flashing green or yellow. It is a sign of air in the system/chip or a potential clog will occur.
            1. At this point stop sorting and get the drops to stabilize
            2. Under the cytometer tab click on the de-bubble function for the chip

3. Under the cleaning section run a bleach cleaning and a di-rinse

5. When you Clog
    a. When you get a partial or full clog in the sort chip. Always remember your sample and your collection tube from the chamber before starting these steps.
        i. Note: Always double check the plates if there was any salt/liquid build up when trying to remove clogs. Give the plate a wipe down with a kimwipe if necessary.
        ii. Note: Check the sample chamber base for any puddles that could have occurred and clean it up.
            1. Partial Clog
                a. In the cytometer tab in the debubble section, select de-bubble chip.
                b. Not cleared in the cytometer tab in the cleaning section, select bleach cleaning.
                c. Stream still did not recover proceed to the clogged chip steps
            2. Clogged
                a. In the cytometer tab of the experiment click on the icon that look like a toolbox in the settings section.
                b. A window called cytometer settings will open. Selecting Advanced Settings at the bottom of the window
                c. The Advanced Settings window will open. On the right side of the window select the 3rd tab called Pressure Options
                d. Click on Standby. The instrument stream will turn off, when the ready option becomes available. Click on it, this will turn the stream on at a high-pressure rate. This should release the clog in the chip.
                e. Only do this step a couple of times. If the stream still has not recovered replace the chip.
                f. To replace the chip in the experiment tab, in the exchange section click on the icon that looks like the chip.